4Casting – Smart Environmental Monitoring System

Computer-Based GUI Environmental Monitoring Prototype

Project Domain: Internet of Things (IoT), Embedded Systems, and Environmental Monitoring

Developed During: Mackethon – Pre-event of Prominence, Department Tech Fest

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Abstract

Environmental monitoring plays an important role in understanding and assessing surrounding environmental conditions. This project presents a Smart Environmental Monitoring System that measures important environmental parameters, including temperature, humidity, light intensity, and air quality, using an Arduino Uno and multiple sensors.

The system uses a DHT11 sensor for temperature and humidity measurement, an LDR for light intensity detection, and an MQ135 gas sensor module for air-quality-related readings. An ESP8266 Wi-Fi module is included for wireless communication. The collected sensor readings are displayed on a 16×2 LCD and transmitted through serial communication for remote data handling.

A computer-based graphical user interface (GUI), developed using Python and Tkinter, provides a visual platform for monitoring environmental parameters. MQTT communication is incorporated into the GUI for receiving sensor data. The system also includes LEDs and a buzzer to indicate different environmental conditions.

The project demonstrates the integration of embedded hardware, sensor interfacing, wireless communication, and computer-based monitoring in an IoT-oriented environmental monitoring application.

1 Introduction

Environmental conditions such as temperature, humidity, light intensity, and air quality influence human health, comfort, and surrounding ecosystems. Monitoring these parameters helps identify changes in environmental conditions and provides information that can support timely action.

Traditional monitoring methods may require manual observation and individual measurement instruments. An automated monitoring system can simplify this process by collecting sensor readings and presenting them through a centralized interface.

The Smart Environmental Monitor was developed to provide a simple and interactive solution for monitoring multiple environmental parameters. The system combines an Arduino Uno with different sensors to collect environmental data. The readings are displayed locally using an LCD and are communicated wirelessly through an ESP8266 module.

A Python-based GUI called **4Casting Station** provides a computer interface for displaying readings, indicating environmental status, and maintaining data logs.

2 Problem Statement

Monitoring multiple environmental parameters independently can be time-consuming and inconvenient. There is a need for a system that can collect environmental readings automatically, display them in an understandable format, and provide alerts when predefined conditions are exceeded.

This project addresses the need for an integrated environmental monitoring prototype using low-cost electronic components and a computer-based graphical interface.

3 Objectives

1. To measure temperature and humidity using a DHT11 sensor.
2. To monitor light intensity using an LDR.
3. To obtain air-quality-related readings using an MQ135 sensor module.
4. To interface multiple sensors with an Arduino Uno.
5. To display environmental readings on a 16×2 LCD.
6. To implement alert indications using LEDs and a buzzer.
7. To incorporate ESP8266-based wireless communication.
8. To develop a Python Tkinter GUI for visualizing environmental data.
9. To maintain a record of received environmental readings.

4 Hardware Components

- **4.1 Arduino Uno:** Main microcontroller board used to acquire sensor readings and control output devices. Processes inputs, updates the LCD, controls LEDs/buzzer, and transmits formatted readings via serial communication.
- **4.2 DHT11 Temperature and Humidity Sensor:** Digital sensor used to measure ambient temperature and relative humidity via a single digital data connection connected to digital pin D2.
- **4.3 Light Dependent Resistor (LDR):** Photoresistor whose resistance varies with incident light intensity. Used to detect surrounding light changes via a voltage-divider connected to analog pin A1.
- **4.4 MQ135 Gas Sensor Module:** Gas sensor used for air-quality-related gas detection, connected to analog pin A2. *Note: Raw analog readings are not calibrated ppm measurements.*
- **4.5 ESP8266 Wi-Fi Module:** Wi-Fi-enabled module used for wireless communication. Receives sensor readings from Arduino via SoftwareSerial at 9600 baud.
- **4.6** 16×2 **LCD Display:** Provides local visual feedback for temperature, humidity, light intensity, and air quality, interfaced in 4-bit mode.
- **4.7 LEDs and Buzzer:** Three LEDs represent environmental statuses (Green: Normal, Yellow: Warning, Red: Critical). A buzzer provides audible alerts during critical conditions.
- **4.8 Breadboard and Connecting Wires:** Used for establishing circuit connections without a permanent PCB.

5 Software Requirements

- **Arduino IDE:** Microcontroller programming and uploading.
- **Embedded C/C++:** Arduino program development.
- **Python:** Computer-side application development.
- **Tkinter:** Graphical user interface development.
- **Paho-MQTT:** MQTT communication package in Python.
- **MQTT Broker:** Message communication using the HiveMQ public broker.

6 System Architecture

The system consists of two major sections:

1. **Hardware Monitoring Unit:** The Arduino Uno receives environmental data from sensors, processes readings, displays them on the LCD, and controls alert indicators while transmitting data to the ESP8266.
2. **Computer-Based Monitoring Unit:** The Python Tkinter application receives data through MQTT, displays environmental parameters, and maintains timestamped logs.

Environmental Sensors $\rightarrow$ *Arduino Uno* $\rightarrow$ *ESP8266* $\rightarrow$ *MQTT Broker* $\rightarrow$ *Python Tkinter GUI*

7 Working Methodology

1. **Data Acquisition:** Arduino collects readings from DHT11, LDR, and MQ135.
2. **Data Processing:** Arduino formats sensor values into a comma-separated string (*Temperature, Humidity, Light Intensity, Air Quality*).
3. **LCD Display:** Real-time local monitoring displayed on 16×2 LCD.
4. **Alert Generation:** Threshold comparisons trigger Green (Normal), Yellow (Warning), or Red + Buzzer (Critical).
5. **Serial Communication:** Formatted data sent to ESP8266 via SoftwareSerial.
6. **Computer-Based Monitoring:** Python Tkinter GUI subscribes via MQTT and logs readings with timestamps. Fallback sample data is generated if live data is unavailable for > 5 seconds.

8 Features

1. Multi-parameter environmental monitoring (Temperature, Humidity, Light, Air Quality).
2. Local 16×2 LCD display and LED/Buzzer alert indications.
3. Serial communication with ESP8266 and MQTT-based Python GUI reception.
4. Timestamped environmental data logging, theme switching, and log viewing.

9 Results and Discussion

The prototype successfully interfaces multiple environmental sensors with the Arduino Uno. Sensor readings are displayed via LCD and Serial Monitor, and status messages are generated based on programmed thresholds. The Python GUI successfully visualizes all parameters, providing status indicators, weather banners, data logging, and theme switching.

10 What We Learned

Through this project, the team gained practical experience in:

- Interfacing multiple analog and digital sensors with Arduino Uno.
- Implementing SoftwareSerial and ESP8266 wireless communication.
- Developing computer-based GUIs in Python using Tkinter and MQTT.
- Designing threshold-based alert mechanisms and hardware-software debugging.

11 Limitations

- DHT11 offers limited accuracy compared to advanced sensors.
- MQ135 analog readings require calibration for precise ppm interpretation.
- Fixed threshold values may not suit every environmental context.
- GUI falls back to sample data if live MQTT data is unavailable.

12 Future Scope

1. Integration of dedicated ESP8266 firmware for robust MQTT publishing.
2. Cloud-based data storage and historical visualization (graphs/charts).
3. Mobile application integration and automated email/SMS alerts.
4. Improved sensor calibration and battery-powered portable operation.

13 Conclusion

The Smart Environmental Monitor successfully demonstrates an embedded monitoring prototype using Arduino Uno, multiple sensors, and a Python-based GUI. The system combines sensor interfacing, local display, alert generation, and remote visualization, offering valuable practical exposure to IoT engineering principles.

14 Project Details

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Event: Mackethon – Pre-event of Prominence, Department Tech Fest
Technologies: Arduino Uno, ESP8266, DHT11, LDR, MQ135, Python, Tkinter, MQTT